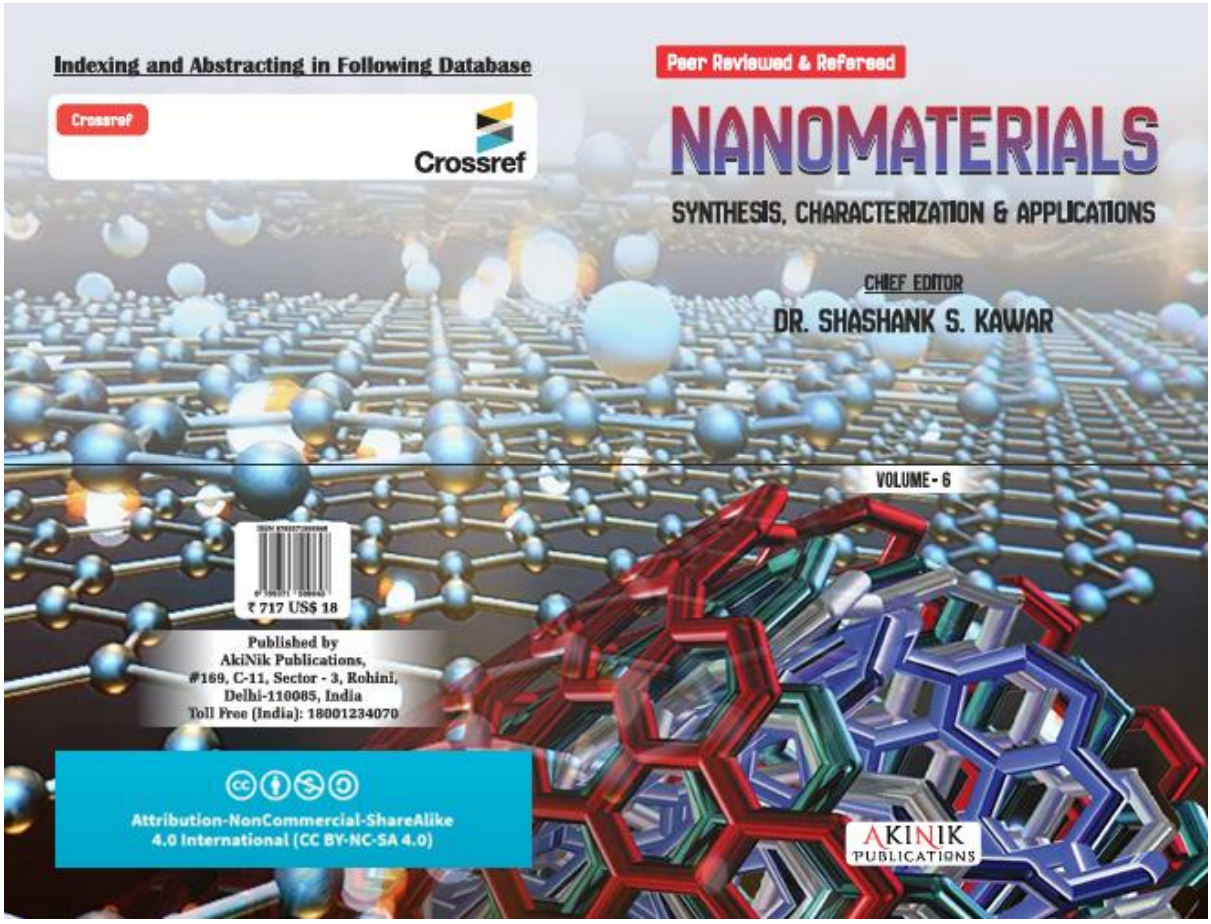
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Chapter - 7
Fabrication and Characterisation of Free-Standing Porous and Non-Porous PCL/PCL-PEO Films using Spin Coating Technique

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Chapter - 7

Fabrication and Characterisation of Free-Standing Porous and Non-Porous PCL/PCL-PEO Films using Spin Coating Technique

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Abstract

Polycaprolactone (PCL) is one of the most extensively investigated biodegradable polyesters in biomedical engineering, valued for its mechanical flexibility, slow hydrolytic degradation over two to four years, and biocompatibility. However, its pronounced hydrophobicity characterised by water contact angles exceeding 80° substantially limits protein adsorption, cell adhesion, and biological integration. Blending PCL with the hydrophilic, water-soluble polymer polyethylene oxide (PEO) provides a straightforward and highly effective strategy to overcome these limitations without altering the fundamental biocompatibility of the system. This chapter presents a systematic and comprehensive investigation into the fabrication of free-standing porous and non-porous thin films from PCL and PCL-PEO blends using spin coating as the primary deposition technique. Four blend compositions (0, 10, 20, and 30 wt% PEO) were prepared and deposited at three spin speeds (500, 1000, and 1500 rpm), generating a 4 × 3 full factorial experimental design comprising 12 unique film conditions tested in triplicate (36 total samples). Surface wettability was quantified by static water contact angle goniometry, and in-vitro degradation was evaluated by monitoring mass loss over 12 weeks in phosphate-buffered saline (PBS, pH 7.4, 37 °C) with periodic SEM and FTIR-ATR analyses. Results confirm that increasing PEO content systematically reduces PCL crystallinity from 57.8% to 34.9%, decreases water contact angle from 82.4° to 47.8°, induces interconnected pore formation with diameters of 2-10 µm, and accelerates in-vitro degradation from 2.1% to 27.6% mass loss at 12 weeks. Film thickness is precisely governed by spin speed, decreasing from approximately 5 µm at 500 rpm to 2 µm at 1500 rpm. These tuneable and interrelated properties establish PCL-PEO spin-coated free-standing thin films as versatile, scalable